

29 Beyond Perturbation Theory

29.4 Supersymmetry Breaking in Gauge Theories

We now turn to a question of great physical interest: in what gauge theories is supersymmetry spontaneously broken?

Let us start with an Abelian gauge theory, the supersymmetric version of quantum electrodynamics described in Section 27.5. This is a $U(1)$ gauge theory with two chiral superfields $\Phi_{\pm}$ carrying $U(1)$ quantum numbers $\pm e$, and superpotential $f(\Phi) = m\Phi_+\Phi_-$. We saw in Section 27.5 that supersymmetry is broken in the tree approximation if we include a Fayet–Iliopoulos term $\xi[V]_D$ in the Lagrangian density, so the Witten index is zero for $\xi \neq 0$ and small e , and hence also for all values of e and ξ , including $\xi = 0$. Is supersymmetry unbroken for $\xi = 0$? It is unbroken in the tree approximation, but how can we tell if higher-order corrections or non-perturbative effects give the vacuum a finite energy in this case?

To answer this we shall use a symmetry principle of the theory for $\xi = 0$, in the way that was described in general terms in Section 29.1. The symmetry here is charge conjugation: the whole Lagrangian density is invariant under the charge conjugation transformation of the chiral and gauge superfields:

$$C\Phi_{\pm}C^{-1} = \Phi_{\mp}, \quad CVC^{-1} = -V. \quad (29.4.1)$$

There is a massless fermion in the tree approximation, the photino, but if we take the vacuum to be even under C then the one-photino state is odd under C , so these states are not related to each other by multiplication with the supersymmetry generator. The chiral fermion here has mass m in the tree approximation, so for small e it is not paired with the vacuum either. With no fermionic state available to pair with, the vacuum state must have strictly zero energy, at least for e small enough so that perturbation theory gives a good qualitative picture of the spectrum. As we saw in Section 29.1, the zero energy of the vacuum implies that supersymmetry is not broken. Likewise, the photino must be strictly massless, since it has no bosonic state with which to pair.

Now, what about values of e that are so large that perturbation theory cannot be trusted at all? The Witten index itself is no help here, because it vanishes. Instead, let us consider the weighted Witten index, $\text{Tr } C(-1)^F$. We have seen that for small e and $\xi = 0$ the vacuum has zero energy, and there are also two zero-energy states containing a zero-momentum photino of spin up or down, and a bosonic zero-energy state containing two zero-momentum photinos of opposite spin. The vacuum makes a contribution $+1$ to $\text{Tr } C(-1)^F$; the two one-photino states make a contribution $+2$ (because both C and $(-1)^F$ are -1); and the two-photino state makes a contribution $+1$, giving a weighted Witten index $\text{Tr } C(-1)^F = 4$. This is independent of the value of e , so even for strong couplings the weighted Witten index is 4, and so supersymmetry is not broken.

There is a complication here [2]. In counting zero-momentum states in the tree approximation, we have not considered the zero-momentum components of the gauge field $V_{\mu}(x)$. A constant term in $V_0(x)$ is no problem, because it can be removed by a gauge

transformation

$$V_\mu(x) \longrightarrow V_\mu(x) + \partial_\mu \Lambda(x), \quad (29.4.2)$$

with gauge parameter $\Lambda(x)$ proportional to x^0 . On the other hand, we cannot simply remove a constant term in $V_i(x)$, because this would require a gauge transformation with $\Lambda(x)$ proportional to x^i , which would conflict with the assumed periodicity of the fields under translations by the box dimension $L \equiv V^{1/3}$. In the particular model under consideration here all fields have charges $\pm e$ or zero, so this periodicity is preserved if we limit ourselves to a lattice of gauge transformations with

$$\Lambda(x) = \frac{2\pi}{eL} \sum_i \ell_i x^i, \quad (29.4.3)$$

where ℓ_i are three positive or negative integers. Therefore although we cannot remove the zero-momentum components of $V_i(x)$ by a gauge transformation, we can freely shift them by amounts $2\pi\ell_i/eL$. The tree-approximation Lagrangian (*not* Lagrangian density) for the x^i -independent part of V_μ in a gauge with $V_0 = 0$ is simply $-\frac{1}{2}L^3 \sum_i (\partial_0 V_i)^2$, so the Hamiltonian is $+\frac{1}{2}L^{-3} \sum_i (\pi_i)^2$, where π_i is the canonical conjugate to V_i : $\pi_i = L^3 \partial_0 V_i$. The wave function $\Psi(V)$ for this field is then just like that of a free particle of unit mass in three dimensions, in a box of linear dimensions $2\pi/eL$ with periodic boundary conditions. The wave functions of definite energy are proportional to $\exp(i\mathbf{k} \cdot \mathbf{V})$, with energy $\mathbf{k}^2/2L^3$, and $k_i = eL\ell_i$, with ℓ_i integers. There is a unique zero-energy state of this field, with $k_i = 0$ and with a normalized wave function equal to the constant $(eL/2\pi)^{3/2}$. Because this state is unique, our counting of zero-energy states is unaffected by the gauge degree of freedom, and the weighted Witten index is indeed 4.

Now let us consider the theory of a simple non-Abelian gauge superfield, without chiral superfields. Witten's 1982 paper [2] on the Witten index presented an argument that for such theories this index is $r + 1$ (or possibly $-r - 1$), where r is the rank of the gauge group, the maximum number of commuting generators. In 1997 he found a correction to this calculation [6], with the result that for the classical unitary, orthogonal, and symplectic groups the index is a Casimir invariant C_1 , which for the unitary and symplectic groups is indeed equal to $r + 1$, but for the orthogonal groups $O(N)$ with $N > 7$ and the exceptional groups takes a different value. In general, this Casimir invariant is defined by Eq. (17.5.33):

$$\sum_{CD} C_{ACD} C_{BCD} = g^2 C_1 \delta_{AB}, \quad (29.4.4)$$

where g is a coupling constant, whose definition can be made unambiguous by specifying that when the generators t_A , t_B , and t_C are restricted to the 'standard' $SU(2)$ subalgebra that was used in the calculation of instanton effects in Section 23.5, the structure constant is simply $g\epsilon_{ABC}$. For the classical groups, we have

$$C_1 = \begin{cases} N & SU(N) \\ N - 2 & SO(N) \text{ for } N > 3 \\ N + 1 & USp(2N) \end{cases} \quad (29.4.5)$$

For $SU(N)$ the rank is $r = N - 1$, while for $USp(2N)$ it is $r = N$, so in both cases $r + 1 = C_1$. But $SO(N)$ with $N > 6$ has rank $r = (N - 1)/2 = (C_1 + 1)/2$ for N odd and $r = N/2 = (C_1 + 2)/2$ for N even, and the index is C_1 , not $r + 1$. Of course, this does not affect the main conclusion that, since the Witten index is not zero, supersymmetry is not spontaneously broken. Kac and Smilga showed in 1999 that the index is also equal to C_1 for exceptional groups [7]. We shall calculate the Witten index here only for $SU(N)$ and $USp(2N)$ supersymmetric gauge theories with no chiral superfields, but in the course of this calculation we will also see why the orthogonal and exceptional groups present special difficulties.

The general strategy of this calculation is the same as for the Abelian theories considered previously. We first examine the states of zero energy, to see if there are any that cannot be paired by action of the supersymmetry generator. If there are, then as long as the coupling is weak enough so that the tree approximation gives a good qualitative picture of the spectrum, we know that these unpaired states really have precisely zero energy. We can then find some non-zero weighted Witten index, which will be constant even for stronger couplings, and conclude that supersymmetry is not broken for any coupling strength.

There is another complication here, that did not enter in the earlier example of an Abelian theory with charged chiral superfields. As usually formulated, general renormalizable theories of gauge bosons and gauginos with no chiral superfields do not contain any dimensionless parameters at all, so that there is no coupling parameter that can be adjusted to make the couplings weak. Instead we have a running coupling constant, depending on the ratio of the energy to a characteristic energy scale Λ , like the scale $\Lambda \approx 200$ MeV for quantum chromodynamics discussed in Section 18.7. For non-Abelian theories the gauge coupling constant becomes strong at energies below Λ , however weak it may be at higher energies. But here we are working in a box of volume L^3 , which provides an infrared cut-off at an energy $\approx 1/L$, that is normally lacking in theories with unbroken gauge symmetries. When we speak of a weakly coupled gauge theory, it should be understood that we mean one with a coupling that is small down to energies of order $1/L$. In this case, it is essential to be able to argue that supersymmetry is broken for strong as well as weak couplings, in order to be able to draw any conclusions at all about the realistic limit of infinite volume.

We will work in temporal gauge, where $V_A^0 = 0$. The Lagrangian density (27.3.1) is then

$$\mathcal{L} = -\frac{1}{4} \sum_{Aij} f_{Aij}^2 - \frac{1}{2} \sum_A (\partial_0 V_{Ai})^2 - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{1}{2} \sum_A D_A^2, \quad (29.4.6)$$

where, in temporal gauge,

$$f_{Aij} = \partial_i V_{Aj} - \partial_j V_{Ai} + \sum_{BC} C_{ABC} V_{Bi} V_{Cj}, \quad (29.4.7)$$

$$(D_i \lambda)_A = \partial_i \lambda_A + \sum_{BC} C_{ABC} V_{Bi} \lambda_C, \quad (29.4.8)$$

$$(D_0 \lambda)_A = \partial_0 \lambda_A. \quad (29.4.9)$$

(As usual, the gauge coupling constant or constants are included as factors in the structure constants C_{ABC} .) Without chiral superfields, there is no other dependence on the auxiliary field D_A ; since it enters quadratically, it may be put equal to the value at which the Lagrangian is stationary, $D_A = 0$, and ignored from now on. Retaining only the x -independent modes, the effective Lagrangian becomes

$$\int d^3x \mathcal{L} = L^3 \left[-\frac{1}{4} \sum_{Aij} \left(\sum_{BC} C_{ABC} V_{Bi} V_{Cj} \right)^2 - \frac{1}{2} \sum_{Ai} (\partial_0 V_{Ai})^2 - \frac{1}{2} \sum_{ABCi} C_{ABC} (\bar{\lambda}_A \gamma_i V_{Bi} \lambda_C) - \frac{1}{2} \sum_A \bar{\lambda}_A \gamma^0 \partial_0 \lambda_A \right]. \quad (29.4.10)$$

The Hamiltonian is then

$$H = \frac{1}{2L^3} \sum_{Ai} \pi_{Ai}^2 + \frac{L^3}{4} \sum_{Aij} \left(\sum_{BC} C_{ABC} V_{Bi} V_{Cj} \right)^2 + \frac{L^3}{2} \sum_{ABCi} C_{ABC} (\bar{\lambda}_A \gamma_i V_{Bi} \lambda_C), \quad (29.4.11)$$

where $\pi_{Ai} = L^3 \partial_0 V_{Ai}$ is the canonical conjugate to V_{Ai} .

The gauge field configurations with zero energy in the tree approximation are those for which $\sum_{BC} C_{ABC} V_{Bi} V_{Cj} = 0$ for all A , i , and j . This condition is always satisfied if V_{Bi} vanishes for all i except where t_B is in a Cartan subalgebra of the gauge Lie algebra.¹ For the unitary and symplectic groups and their direct products this is the only way that it can be satisfied. The same is true for the orthogonal gauge algebras $O(N)$ with $N \leq 6$, which are all equivalent to symplectic and/or unitary Lie algebras (see the Appendix to Chapter 15), but not for orthogonal gauge algebras $O(N)$ with $N \geq 7$, which is why Witten's original calculation needed correction for this case.

In the rest of this calculation we will only consider gauge algebras like the symplectic and unitary algebras for which the condition of zero energy in the tree approximation does require the V_{Bi} to vanish for all i except where t_B is in a Cartan subalgebra of the gauge Lie algebra. With all $V_{Ai} = 0$ except for the V_{Ai} for which t_A is in the Cartan subalgebra, the zero-energy modes of the fermion field are those also with $\lambda_A = 0$, except for the λ_A for which t_A is in the Cartan subalgebra.

Now we must count these states. The eigenvalues of the t_A in any representation of a semi-simple Lie algebra are quantized, so by gauge transformations all values of the non-zero gauge fields V_{Ai} are equivalent to values in a finite box with periodic boundary conditions. The quantization of these modes is just like that carried out for the gauge field

¹A Cartan subalgebra is any subalgebra spanned by r -independent generators t_A that commute with one another, that is, for which C_{AAB} vanishes for all t_A when t_A and t_B are in the Cartan subalgebra, where r is the rank, the maximum number of such generators. For instance, for the $SU(3)$ symmetry of strong interactions discussed in Section 19.7, the rank is $r = 2$, and the Cartan subalgebra can be taken to consist of the third component of isospin t_3 and the hypercharge t_8 , which act on the light quarks with the commuting matrices denoted λ_3 and λ_8 in Eq. (19.7.2).

in the $U(1)$ model considered above, so the zero-energy state of these fields is again unique, with a constant wave function in the box.

The multiplicity of states with zero energy in the tree approximation comes entirely from the fermion degrees of freedom. It is convenient to use a two-component notation, in which instead of a four-component Majorana field $\lambda_{A\alpha}$ for each generator of the Cartan subalgebra, we have two left-handed fields λ_{ALa} with $a = \pm 1/2$, and their right-handed Hermitian adjoints $\lambda_{ALa}^\dagger$. These gaugino fields satisfy the canonical anticommutation relations

$$\{\lambda_{ALa}, \lambda_{BLb}^\dagger\} = \delta_{AB}\delta_{ab}$$

and

$$\{\lambda_{ALa}, \lambda_{BLb}\} = \{\lambda_{ALa}^\dagger, \lambda_{BLb}^\dagger\} = 0.$$

By operating on an arbitrary state vector with as many factors of λ_{ALa} as necessary, we can construct a state vector $||0\rangle$ which is annihilated by all λ_{ALa} . The general zero-energy state vector is then a linear combination of products of the $\lambda_{ALa}^\dagger$ acting on $||0\rangle$.

To see which of these states may be paired with each other by the action of the supersymmetry generator, we must take account of a symmetry of the theory. The zero-energy condition, that the gauge and gaugino fields lie only in directions corresponding to the Cartan subalgebra, is invariant under the subgroup of the original gauge group consisting of elements h that leaves this subalgebra invariant, that is, for which $h^{-1}t_A h$ is a linear combination of the t_B . These form a finite group, known as the Weyl group. For instance, in the defining representation of $SU(N)$ the Weyl group consists of permutations of the N coordinate axes, together with multiplication by a phase needed to make the transformations unimodular. These can be represented by products of the finite gauge transformations $W(i, j) = \exp(i\pi\sigma(ij)/2) = i\sigma(ij)$ with $i \neq j$ that permute the i th and j th coordinate axes, where $\sigma(ij)$ is the $U(N)$ generator with $[\sigma(ij)]^i_j = [\sigma(ij)]^j_i = 1$, and with all other elements zero. These transformations induce orthogonal transformations in the space spanned by the diagonal traceless Hermitian matrices t_A that generate the Cartan subalgebra in the adjoint representation. For instance, for the group $SU(2)$ we can take the Cartan subalgebra to consist of just t_3 , and the Weyl group then consists of the unit element and a single non-trivial gauge transformation $W(1, 2) = i\sigma(1, 2) = i\sigma_1$, for which $W^{-1}t_3W = -t_3$. For $SU(3)$ the Cartan subalgebra has the two generators λ_3 and λ_8 , and the Weyl group consists of the six gauge transformations 1 , $W(1, 2)$, $W(2, 3)$, $W(1, 3)$, $W(1, 2)W(2, 3)$, and $W(2, 3)W(1, 2)$, which generate rotations by multiples of 60° in the space spanned by t_3 and t_8 .

Assuming the vacuum to be invariant under the Weyl group, it can be paired by action of the supersymmetry generators only with other states that are invariant under the Weyl group.² This may or may not include the previously constructed zero-energy state $||0\rangle$. The condition that the state $||0\rangle$ be annihilated by all λ_{ALa} is obviously invariant under the Weyl group, so if it is unique this state must furnish a one-dimensional representation of

²Witten remarked that physical states are necessarily Weyl-invariant, but we will not need to go into this here, because the Weyl invariance of the vacuum and of the supersymmetry generators means that only Weyl-invariant states are relevant to spontaneous supersymmetry breaking.

the Weyl group. The Weyl group always acts on the generators of the Cartan subalgebra by orthogonal transformations, so there are two such representations: the invariant representation, in which each Weyl transformation is represented by unity, and the pseudoinvariant representation, in which each Weyl transformation is represented by the determinant of its action on the generators of the Cartan subalgebra.

Let us first consider the case where $\|0\rangle$ is invariant under the Weyl group. Obviously no linear combination of the one-fermion states $\lambda_{ALa}^\dagger \|0\rangle$ can be invariant under the Weyl group. There is just one Weyl-invariant linear combination of the two-fermion states; it is of the form $U\|0\rangle$, where

$$U \equiv \sum_{abA} e_{ab} \lambda_{ALa}^\dagger \lambda_{ALb}^\dagger. \quad (29.4.12)$$

(The spin indices a and b are contracted with the antisymmetric tensor e_{ab} defined by Eq. (25.2.9), because the anticommutation relations make the product $\lambda_{ALa}^\dagger \lambda_{ALb}^\dagger$ antisymmetric in a and b .) There are various Weyl-invariant linear combinations of products of three or more generators of the Cartan subalgebra, but the anticommutation of the $\lambda_{ALa}^\dagger$ makes all of them vanish³ except for powers of U . Also, in the product of more than r of the U s some of the $\lambda_{ALa}^\dagger$ would have to appear twice, so $U^{r+1} = 0$. We conclude then that the Weyl-invariant states are limited to the $r + 1$ states

$$\|0\rangle, \quad U\|0\rangle, \quad U^2\|0\rangle, \quad \dots, \quad U^r\|0\rangle. \quad (29.4.13)$$

These are all bosonic states, and there are no fermionic states with zero energy in the tree approximation with which they could be paired, so for sufficiently weak coupling these states must have precisely zero energy, and supersymmetry is unbroken. Also, the Weyl-invariant Witten index here is $r + 1$, and this is independent of the coupling strength, so *supersymmetry is not spontaneously broken whatever the strength of the gauge coupling*.

³To see this for $SU(N)$, note that each generator t_A of its Cartan subalgebra may be written as a linear combination $t_A = \sum_i c_{Ai} T_i$ of generators T_i of the Cartan subalgebra of $U(N)$, with the only non-zero element of each T_i being $(T_i)^i_i = 1$ (with indices not summed). In order for the t_A to be traceless, we must have $\sum_i c_{Ai} = 0$. Since the fields $\lambda_{ALa}^\dagger$ transform under the Weyl group like the t_A , it follows that for a function $\sum_{AB\dots} d_{AB\dots} \lambda_{ALa}^\dagger \lambda_{BLb}^\dagger \dots$ to be Weyl-invariant, the coefficients $d_{AB\dots}$ must take the form

$$d_{AB\dots} = \sum_{ij\dots} c_{Ai} c_{Bj} \dots D_{ij\dots},$$

where the $D_{ij\dots}$ are invariant tensors, in the sense that for any vectors u_i, v_j , etc., the function $D(u, v, \dots) \equiv \sum_{ij\dots} D_{ij\dots} u_i v_j \dots$ is invariant under permutations of the coordinate axes. The most general such function is a linear combination of products of the function

$$S(x, y, z, \dots) = \sum_i x_i y_i z_i \dots,$$

with arguments $x, y, z, \dots$ taken as various subsets of the u, v , etc. But because $\sum_i c_{Ai} = 0$, in our case the sum of the components of each vector vanishes, so $D(u) = S(u) = 0$; $D(u, v)$ is proportional to $S(u, v)$; $D(u, v, w)$ is proportional to $S(u, v, w)$; $D(u, v, w, x)$ is a linear combination of $S(u, v, w, x)$, $S(u, v)S(w, x)$, $S(u, w)S(v, x)$, and $S(u, x)S(v, w)$; and so on. The important point is that, even though the function $D(u, v, \dots)$ may not be symmetric in its arguments (because different products of S s may appear with different coefficients), the functions $S(u, v, \dots)$ are symmetric. In our case, the vectors are the anticommuting quantities $u(a)_i = \sum_A \lambda_{ALa}^\dagger c_{Ai}$, for which the only non-vanishing S function is $S(u(1/2), u(-1/2))$. With a suitable normalization of the generators, this is the operator U .

In the case where the state $\|0\rangle$ is pseudoinvariant, the only Weyl-invariant states are of the form

$$\sum_{\mathcal{A}, \mathcal{B}, \dots} \epsilon_{\mathcal{A}, \mathcal{B}, \dots} \lambda_{\mathcal{A}La}^\dagger \lambda_{\mathcal{B}Lb}^\dagger \cdots \|0\rangle, \quad (29.4.14)$$

where $\epsilon_{\mathcal{A}, \mathcal{B}, \dots}$ is the totally antisymmetric tensor of rank r . This is completely symmetric among the $a, b, \dots$, so each state is characterized by the number of these indices that are $+1/2$ rather than $-1/2$, a number that can take any value from 0 to r , so the number of independent states is $r+1$. Depending on whether r is even or odd, these states are either all bosonic or all fermionic, so the Witten index here is $\pm(r+1)$, and again supersymmetry is not spontaneously broken.

There is an interesting relation between the value of the Witten index obtained here and ideas of how certain global symmetries become broken. The Lagrangian for a supersymmetric gauge theory is invariant under the transformations of a global $U(1)$ ‘ R -symmetry,’ which change the left- and right-handed parts of the gaugino fields by opposite phases:

$$\lambda_{AL} \longrightarrow e^{i\varphi} \lambda_{AL}, \quad \lambda_{AR} \longrightarrow e^{-i\varphi} \lambda_{AR}, \quad (29.4.15)$$

with φ an arbitrary real constant phase. The conservation of the current J_5^μ associated with this symmetry is violated by an anomaly

$$\partial_\mu J_5^\mu = -\frac{1}{32\pi^2} \sum_{ABCD} C_{ACD} C_{BCD} f_A^{\mu\nu} f_B^{\rho\sigma} \epsilon_{\mu\nu\rho\sigma}, \quad (29.4.16)$$

where as usual $\epsilon_{\mu\nu\rho\sigma}$ is the totally antisymmetric quantity with $\epsilon^{0123} \equiv 1$. (This is obtained from Eq. (22.2.24) by taking the gauge group generators t_A as $(t_A)_{BC} = -iC_{ABC}$, because the gauginos are in the adjoint representation of the gauge group, and by multiplying the anomaly with a factor $1/2$, because the gauginos do not have distinct antiparticles.) With the definition of the gauge coupling specified above, Eq. (23.5.20) gives the integral of the product of field strengths in the anomaly as

$$\epsilon_{\mu\nu\rho\sigma} \int d^4x \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} = 64\pi^2 \nu / g^2, \quad (29.4.17)$$

where the ‘winding number’ ν is an integer characterizing the topological class to which the gauge field belongs. Putting together Eqs. (29.4.4), (29.4.16), and (29.4.17), we see that an instanton of winding number ν induces a change in $R \equiv \int d^3x J_5^0$ given by

$$\Delta R = \int d^4x \partial_\mu J_5^\mu = -2\nu C_1. \quad (29.4.18)$$

That is, the effective action contains terms $(\sum_{Aab} \lambda_{ALa} \lambda_{ALb} e_{ab})^{C_1}$ and its integer powers that, instead of being invariant under the R -symmetry transformations (29.4.15), are transformed by integer powers of the phase $\exp(2i\varphi C_1)$. Thus instantons invalidate invariance under the general $U(1)$ R -transformation (29.4.15), reducing it to the group $\mathbb{Z}_{2C_1}$ of transformations (29.4.15) with φ an integer multiple of π/C_1 . We might expect that the growth of the gauge coupling at low energy would lead (as in quantum chromodynamics) to the

appearance of vacuum expectation values for gaugino bilinears, which would mean that the discrete symmetry group $\mathbb{Z}_{2C_1}$ is spontaneously broken to its $\mathbb{Z}_2$ subgroup, generated by a simple sign change of the gaugino multiplet. Then there would be C_1 zero-energy states $\|n\rangle$, given by acting on any one vacuum state $\|X\rangle$ with the elements $\exp(in\pi R/C_1)$ of $\mathbb{Z}_{2C_1}$, with n running only over the values $0, 1, \dots, C_1 - 1$ because we treat as equivalent any pair of states that differ only by action of the generator $\exp(i\pi R)$ of $\mathbb{Z}_2$. Since $\|X\rangle$ can only be a linear combination of states with even values of R , we can form C_1 states with all values $\mathcal{R} = 0, 2, \dots, 2C_1 - 2$ of R by taking the linear combinations

$$\sum_n \exp\left(\frac{in\pi\mathcal{R}}{C_1}\right) \|n\rangle.$$

In particular, for $SU(N)$ and $USp(2N)$ these states are the same as the Weyl-invariant states encountered in the calculation of the Witten index in the case where the state $\|0\rangle$ is R -invariant, and Weyl-invariant rather than pseudoinvariant. The operator U defined by Eq. (29.4.12) has $R = 2$, so there are $r + 1 = C_1$ states $U^n\|0\rangle$ with $R = 2n$ running from zero to $2r = 2C_1 - 2$. More generally, the presence of C_1 zero-energy states with the same statistics helps to explain why the Witten index is equal to $\pm C_1$ for the exceptional and orthogonal as well as the unitary and symplectic groups.

The fact that the Witten index is non-vanishing for all pure gauge supersymmetric theories means that to find examples of spontaneous supersymmetry breaking we must add chiral superfields to the theory. It does not help to add *massive* chiral superfields to the theory, since for weak coupling the introduction of massive fields does not change the menu of zero-energy states. We have already seen an example of this: the work of Affleck, Dine, and Seiberg reviewed in Section 29.3 showed that an $SU(N_c)$ gauge theory with $N_f < N_c$ left-chiral superfields Q_{ai} in the defining representation of $SU(N_c)$ and an equal number of left-chiral superfields $\bar{Q}_{ai}$ in the complex conjugate representation, with a mass term $\sum_{aij} m_{ij} \bar{Q}_{ai} Q_{aj}$, has N_c zero-energy bosonic states and no zero-energy fermionic states. (This is one case where the Witten index is not left unchanged when a mass term vanishes, because this mass term is the term in the superpotential of highest order in the superfields, so that a vanishing mass changes the behavior of the superpotential for large superfields. In fact, Eq. (29.3.31) shows that as the masses go to zero the scalar field values in the state of zero energy go to infinity.)

On the other hand, there is no difficulty in finding theories with left-chiral superfields subject to a symmetry that keeps them massless, in which supersymmetry is dynamically broken. For instance, consider an $SU(N_c)$ gauge theory with N_f left-chiral superfields Q_{ai} and N_f left-chiral superfields $\bar{Q}_{ai}$ again in the N_c and $\bar{N}_c$ representations of the gauge group, but now also with N_f left-chiral superfields L_i that are neutral under the gauge group $SU(N_c)$. Assume a global (or weakly coupled local) $SU(N_f)$ symmetry, acting on the ‘flavor’ i indices of the Q s and the L s but not the $\bar{Q}$ s, which among other things forbids a mass term linking the Q s and $\bar{Q}$ s. We take Q_{ai} and L_i in the representations N_f and $\bar{N}_f$ of $SU(N_f)$, respectively, while the $\bar{Q}$ s are taken to be $SU(N_f)$ singlets. The only

renormalizable superpotential is then of the form

$$f(Q, \bar{Q}, L) = \sum_{ija} \mathcal{G}_j \bar{Q}_{aj} Q_{ai} L_i, \quad (29.4.19)$$

where the $\mathcal{G}_j$ are a set of coupling constants, which by an $SU(N_f)$ rotation may be chosen to have only one non-vanishing component, say the one with $j = N_f$, which can also be chosen to be positive. (This is a generalization of a model treated by Affleck, Dine, and Seiberg [8], in which they took $N_c = 3$ and $N_f = 2$.) The gauge-neutral superfields L_i have no effect on the non-perturbative terms in the effective superpotential, so we can use the result of Section 29.3, that for $N_c > N_f$ the gauge interactions yield a total effective superpotential

$$f_{\text{total}}(Q, \bar{Q}, L) = \sum_{ija} \mathcal{G}_j \bar{Q}_{aj} Q_{ai} L_i + \mathcal{K} \left[\text{Det}_{ij} \sum_a \bar{Q}_{aj} Q_{ai} \right]^{-1/(N_c - N_f)}, \quad (29.4.20)$$

with $\mathcal{K}$ a constant. In order for supersymmetry to be unbroken it is necessary (though not sufficient) that the scalar components q_{aj} , $\bar{q}_{aj}$, and ℓ_i of the chiral superfields satisfy the condition $\partial f_{\text{total}}(q, \bar{q}, \ell)/\partial \ell_i = 0$, so that, for all i ,

$$\sum_{ja} \mathcal{G}_j \bar{q}_{aj} q_{ai} = 0. \quad (29.4.21)$$

But this tells us that the matrix $\sum_a \bar{q}_{aj} q_{ai}$ has a zero eigenvalue, and therefore has zero determinant, so this is a singular point of the superpotential (29.4.20), at which it is impossible for $\partial f_{\text{total}}/\partial q_{ai}$ or $\partial f_{\text{total}}/\partial \bar{q}_{aj}$ to vanish. Supersymmetry is therefore necessarily broken in this class of models.

For instance, Affleck, Dine, and Seiberg [8] found the non-vanishing scalar components q_{aj} , $\bar{q}_{aj}$, and ℓ_i of the superfields Φ_{ai} , $\bar{\Phi}_{ai}$, L_i at the minimum of the potential in their model with $N_c = 3$ and $N_f = 2$ to be at the values

$$\begin{aligned} q_{11} &= \bar{q}_{11} = 1.286 (\mathcal{K}/2\mathcal{G})^{1/7}, \\ q_{22} &= \bar{q}_{22} = 1.249 (\mathcal{K}/2\mathcal{G})^{1/7}, \\ \ell_1 &= \sqrt{q_{11}^2 - q_{22}^2}, \end{aligned} \quad (29.4.22)$$

where the vacuum energy density is

$$\rho_{\text{VAC}} = 3.593 (16\mathcal{G}^{10}/\mathcal{K}^4)^{1/7}. \quad (29.4.23)$$

Also, $\mathcal{K} = 2\Lambda^7$ for $\theta = 0$.

The fact that supersymmetry is spontaneously broken in this case encourages us to think that it will be broken by strong gauge forces in a wide range of asymptotically free gauge theories, and thus lends legitimacy to the speculations about supersymmetry breaking in Section 28.3.